

3.3.1 Progressive and stationary waves

3.3.1.1 Progressive waves

Content	Opportunities for skills development
Oscillation of the particles of the medium; amplitude, frequency, wavelength, speed, phase, phase difference, $c = f\lambda$ $f = \frac{1}{T}$ Phase difference may be measured as angles (radians and degrees) or as fractions of a cycle.	PS 2.3 / MS 0.1, 4.7 / AT a, b Laboratory experiment to determine the speed of sound in free air using direct timing or standing waves with a graphical analysis.

3.3.1.2 Longitudinal and transverse waves

Content	Opportunities for skills development
Nature of longitudinal and transverse waves. Examples to include: sound, electromagnetic waves, and waves on a string. Students will be expected to know the direction of displacement of particles/fields relative to the direction of energy propagation and that all electromagnetic waves travel at the same speed in a vacuum. Polarisation as evidence for the nature of transverse waves. Applications of polarisers to include Polaroid material and the alignment of aerials for transmission and reception. Malus's law will not be expected.	PS 2.2, 2.4 / MS 1.2, 3.2, 3.4, 3.5 / AT i Students can investigate the factors that determine the speed of a water wave.

3.3.1.3 Principle of superposition of waves and formation of stationary waves

Content	Opportunities for skills development
Stationary waves. Nodes and antinodes on strings. $f = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$ for first harmonic. The formation of stationary waves by two waves of the same frequency travelling in opposite directions. A graphical explanation of formation of stationary waves will be expected. Stationary waves formed on a string and those produced with microwaves and sound waves should be considered. Stationary waves on strings will be described in terms of harmonics. The terms fundamental (for first harmonic) and overtone will not be used.	MS 4.7 / PS 1.2, 2.1 / AT i Students can investigate the factors that determine the frequency of stationary wave patterns of a stretched string.
Required practical 1: Investigation into the variation of the frequency of stationary waves on a string with length, tension and mass per unit length of the string.	